



V93000 Specifications

V93000 Wave Scale MX Card

Version 7.10.9

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➡ Do not proceed beyond a warning unless the indicated conditions are fully understood and met.

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Wave Scale MX Card

Wave Scale MX is the mixed-signal instrument for the Advantest V93000 test system, requiring SmarTest release 7.10.6 or higher. A single card provides 16x Arbitrary Waveform Generators (AWGs) and 16x Digitizers that can be connected either single-ended or differential.



V93000 SoC Wave Scale MX Digital Card

Contact information

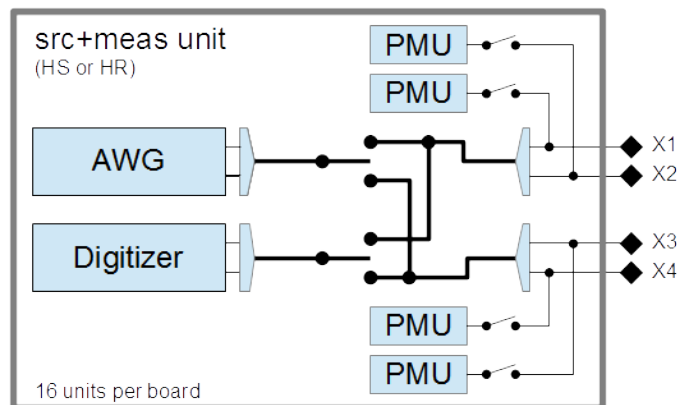
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Wave Scale MX overview

Wave Scale MX is built on the concept of totally independent units. Each unit contains an AWG and a Digitizer that can be flexibly connected to 2 differential pogo pairs. In addition, each pogo can be connected to a parametric measurement unit (PMU) for DC measurements. The figure below shows a block diagram of a single unit.



Wave Scale MX unit block diagram

A Wave Scale MX card always consists of 16 units. However, there are 2 different types of units:

- High-Speed (HS) unit
16bit AWG (220MHz bandwidth) and 16bit Digitizer (300MHz bandwidth)
- High-Resolution (HR) unit
24bit AWG (90kHz bandwidth) and 24bit Digitizer (500kHz bandwidth)

Based on these units, 3 variants of Wave Scale MX are available:

- E9740A/B-OP
Wave Scale MX high-speed and high-resolution contains 8x HS and 8x HR units
- E9740HR-OP
Wave Scale MX high resolution only contains 16x HR units
- E9740HS/HSB-OP
Wave Scale MX high-speed only contains 16x HS units

The AWG and the digitizer of a unit operate independently and fully parallel. The AWG and the digitizer can be connected either single-ended to any of the four pogos X1, X2, X3 or X4, or differential to each pair of pogo pins X1/X2 or X3/X4. It is recommended to route the AWG to the connectors X1 and X2. A PMU is associated with each pogo. With the built-in 2:1 multiplexer the digitizer can be connected to either connector X3 or X4.

The number of units that can be used is controlled by licensing. The E9740A card can be used with the N9740HRBC high resolution base capability license (4 HR units), the N9740HSBC high speed base capability license (4 HS units), or both base capability licenses (4 HR units and 4 HS units), depending on your application.

To be operable, each Wave Scale MX board E9740HR requires a base capability license N9740HRBC, which enables 4 units. Additional units can be enabled by adding the optional N5940U floating licenses.

To be operable, each Wave Scale MX board E9740HS requires a base capability license N9740HSBC, which enables 4 units. Additional units can be enabled by adding the optional N5941U floating licenses. The base license enables the AWG and digitizer with bandwidths up to 30 MHz and the High-Speed digitizer sample rates up to 100 Mbps. For measurements that

require a bandwidth greater than 30 MHz, a bandwidth upgrade license (N5942U) per unit is required. For complete information on the base and extended capabilities of the WSMX card variants, see "Wave Scale MX functionalities and features (Topic 250473)".

Scope of specifications

The Advantest V93000 specifications are valid at the device with an adequate DUT board and valid calibration, i.e. system calibration (see section "Calibration and Operation", below) and fixture delay measurement (TDR). Since design and manufacturing of the DUT board are beyond Advantest's control, Advantest specifies and verifies the specifications up to the pogo level.

Specifications describe warranted product performance.

Characteristics, included as typical values, provide additional useful information, but do not represent warranted performance.

These specifications are valid for recommended system configurations.

These specifications are applicable for SmarTest 7.10.9.

Radiated Immunity

All specifications and typical performance numbers given in this document are valid in environments with an electrical field strength of maximum 3V/m.

Calibration and Operation

Warm up time	60 minutes
Calibration period	3 months
Reference calibration period	6 months

Valid at ambient temperature within $\pm 2.5K$ of calibration temperature. For temperature requirements during calibration, please refer to the "Maintenance Guide".

The external calibration and external adjustment period for a newly shipped system begins on the date of installation at your site, and not on the factory calibration and adjustment date disclosed on the Certificate of Calibration (CoC). As long as the shelf life (two years) is not expired at date of installation, the factory calibration and adjustment is still valid for the subsequent six months. Two years and six months after factory calibration and adjustment, the external calibration and external adjustment is out-of-date in any case.

Environmental

Operating temperature	15°C to 30°C (59°F to 86°F)
Specification warranted temperature	20°C to 30°C (68°F to 86°F)
Maximum humidity at 30°C	<80% R.H., non condensing
Storage (without water)	-40°C to +70°C

Compatibility

Wave Scale MX can be installed in a slot powered by universal DC/DC power units (E8003AA). It is compatible with Smart Scale generation or later digital cards and power supplies for example, PS1600, PS9G, DPS128) but can also be used without any other type of card.

System software

- Wave Scale MX requires SmarTest release 7.10.6 or higher
- Linux Red Hat Enterprise 7 or later releases supported by SmarTest

Wave Scale MX pogo pin assignment and card configuration in the model file

The pogo blocks of the Wave Scale MX card are split into high speed (HS) and high resolution (HR), meaning that there are separate standard locations for the HS and HR units.

There are two pogo pin assemblies available to connect the Wave Scale MX card to the pogo blocks on the test head with associated licensing requirements:

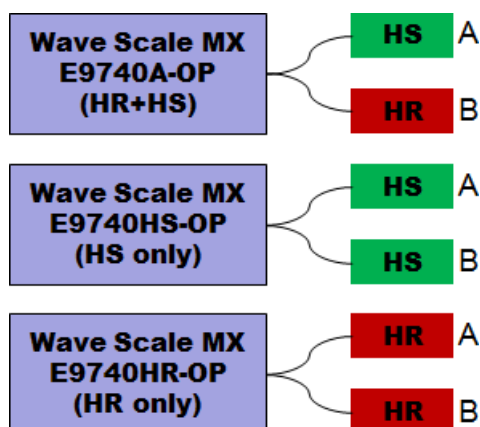
- *High-density pogo pin assembly (2x32 pins)* on page 8
- *Standard-density compatible pogo pin assembly (4x16 pins)* on page 10
- *Licensing* on page 12
- *Device license file* on page 13

High-density pogo pin assembly (2x32 pins)

A high-density pogo pin assembly requires the E9740P pogo cable. This cable connects a Wave Scale MX card to two pogo blocks with 16 differential signals per pogo block. :

- In case of the E9740A/B-OP card variant, one pogo block provides access to eight HR units, the other one provides access to eight HS units.
- In case of the E9740HS/HSB-OP card variant, the two pogo blocks are fully identical and each one provides access to eight HS units.
- In case of the E9740HR-OP card variant, the two pogo blocks are fully identical and each one provides access to eight HR units.

Pogo blocks (high-density)



A: units 1-8
B: units 9-16

Pogo pin mapping (high-density)

For the high-density pogo blocks, existing pogo locations have the same name as the naming for the standard-density pogo blocks. The additional pogo pin locations are derived from the neighboring compatible pogo pin location by adding 400. In the pogo block below, the pins marked "nnn" are the compatible pins, and the pins marked "mmm" are the additional high density pins.



	a	b	c	d
01		1X1 nnn01		
02	5X1 mmm01		1X3 nnn02	
03		1X2 nnn03		5X3 mmm02
04	5X2 mmm03		1X4 nnn04	
05		2X1 nnn05		5X4 mmm04
06	6X1 mmm05		2X3 nnn06	
07		2X2 nnn07		6X3 mmm06
08	6X2 mmm07		2X4 nnn08	
09		3X1 nnn09		6X4 mmm08
10	7X1 mmm09		3X3 nnn10	
11		3X2 nnn11		7X3 mmm10
12	7X2 mmm11		3X4 nnn12	
13		4X1 nnn13		7X4 mmm12
14	8X1 mmm13		4X3 nnn14	
15		4X2 nnn15		8X3 mmm14
16	8X2 mmm15		4X4 nnn16	
17		8X4 mmm16		

●
Orientation mark

nnn : pogo block number
mmm = nnn + 400

Example: nnn = 323
 mmm = 723

standard pogo locations:

HS 1 :	426/826
HS 2 :	320/720
HS 3 :	232/632
HS 4 :	319/719
HR 1 :	323/723
HR 2 :	432/832
HR 3 :	324/724
HR 4 :	427/827

The orientation mark on each pogo block faces towards the inner side of the DUT interface.

Differential pairs (high-density)

The following diagram illustrates the differential pairs of the high-density pogo block A.
For block B, unit numbers 1-8 must correspondingly be replaced with 9-16.

Naming: 1X1 = Unit 1, pogo X1

	a	b	c	d
01		1X1		
02	5X1		1X3	
03		1X2		5X3
04	5X2		1X4	
05		2X1		5X4
06	6X1		2X3	
07		2X2		6X3
08	6X2		2X4	
09		3X1		6X4
10	7X1		3X3	
11		3X2		7X3
12	7X2		3X4	
13		4X1		7X4
14	8X1		4X3	
15		4X2		8X3
16	8X2		4X4	
17		8X4		

differential pair

Orientation mark

WSMX card configuration in the model file (high-density pogo cable)

The following is an example of the `CARD_CONFIGURATION` section in the model file for allocating the high-density pogo blocks for a Wave Scale MX card:

```
CARD_CONFIGURATION
# WSMXA-OP with HD pogo cable
  slot = 101, HW = WSMX, pogoblocks=HS:(231 631) HR:(232 632), type=HD2x32

# WSMXHS-OP with HD pogo cable
  slot = 108, HW=WSMX, pogoblocks=HS:(426 826 320 720), type=HD2x32

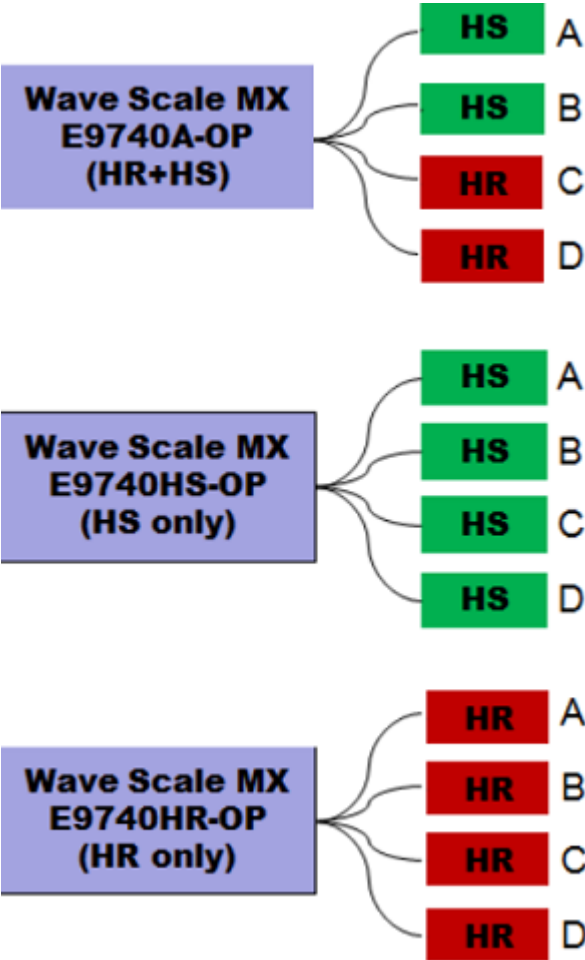
# WSMXHR-OP with HD pogo cable
  slot = 101, HW = WSMX, pogoblocks=HR:(231 631 232 632), type=HD2x32
```

Standard-density compatible pogo pin assembly (4x16 pins)

A standard-density pogo pin assembly requires the pogo cable E9740PSD and provides DUT board compatibility for devices that use E9740A/B-OP, E9740HS/HSB-OP, or E9740HR-OP card variants. A Wave Scale MX card connects to four pogo blocks with 8 differential signals per pogo block.

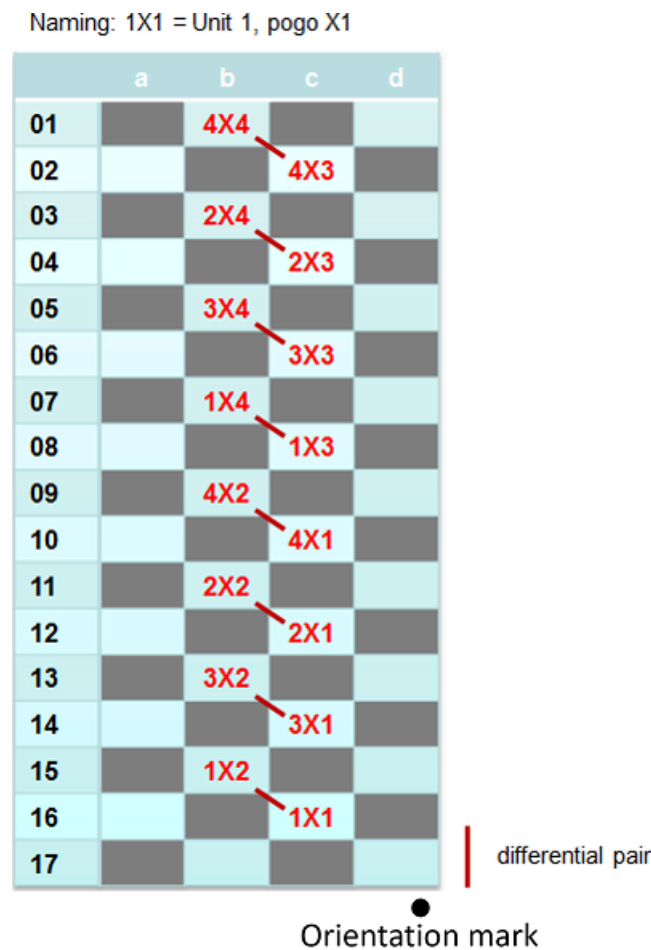
- In case of the E9740A/B-OP card variant, two pogo blocks provide access to two HR units, the other two provide access to two HS units.
- In case of the E9740HS/HSB-OP, the four pogo blocks are fully identical and each one provides access to four HS units.
- In case of the E9740HR-OP card variant, the four pogo blocks are fully identical and each one provides access to four HR units.

Pogo blocks (standard-density)



Pogo pin mapping (standard-density)

The following diagram illustrates the standard-density pogo block layout:



Model file example (standard-density)

This example shows the allocation of four standard compatible pogo blocks "231,431,228,428":

```
CARD_CONFIGURATION

# WSMXHS-OP with SD pogo cable
  slot = 108, HW = WSMX, pogoblocks=HS:(231 431 228 428)
```

Licensing

For a complete list of available licenses for all variants, refer to V93000 License Master Look-up Table (topic 130665) in the SmarTest documentation package and set the filter of the **Main product** column to WSMX.

License checks are carried out during the bind and not reflected in the instrument model. If a license is required but not available, a bind error is reported.

The following licenses are available to enable the full range of Wave Scale MX card features:

Base Capability

Required to activate the Wave Scale MX card (4 HR or HS units per license).

Base Capability Extensions

Upgrades are available for the Base Capability license, adding:

- High resolution
- High-speed with 30 MHz bandwidth

- High-speed with full 500 MHz bandwidth

Device license file

A Wave Scale MX card Base Capability license (high-resolution or high-speed) is required.

Details of the available licenses can be found in V93000 License Master Look-up Table (topic 130665) in the SmarTest documentation package (filter with **Product description = Wave Scale MX** and **DLF/TPLF definition = bandwidth**) for the full bandwidth licenses.

The `bandwidth` parameter is used to specify the bandwidth availability for WSMX-HS cards as follows:.

0MHz - Disabled

30MHz - Base capability

500MHz - Full bandwidth

For enabling higher bandwidth (above 30 MHz), apply the Wave Scale MX-HS high-speed unit upgrade to full bandwidth license and specify the `bandwidth=500MHz` option in the device license file.

Example for configuring the Wave Scale MX licenses in the device license file:

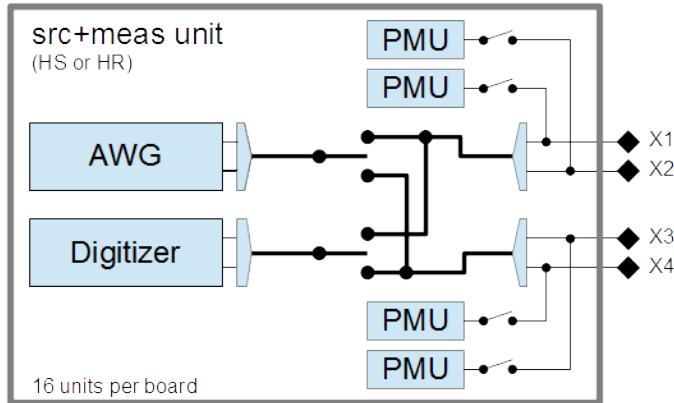
```
...  
  
IOCHANNEL  
  
21801-21808: HW = WSMX-HS, bandwidth = 500MHz  
23201-23208: HW = WSMX-HR  
...  
...
```

For details about the device licensing file, see *Device license file* and *Device license file syntax*.

In general, for release to production, we recommend creating a device license file using the Testflow Analyzer Tool (TAT). However, TAT does not list the bandwidth for WSMX HS cards. This must be added manually into the device license file.

Wave Scale MX unit block diagram

The figure below shows the block diagram of high-resolution (HR) or high-speed (HS) unit.



Wave Scale MX unit block diagram

Within a unit, both PMU, AWG and Digitizer can be connected to the pogos such that

- Each pogo can either be a source or a measure pogo
- Both AWG and Digitizer have a differential 2:1 or a single-ended 4:1 multiplexer
- Each pogo can be connected to an independent PMU

Wave Scale MX parallel usable functionality per unit

The table below shows all combinations that can be used (y) or not (n) in parallel within a single unit. There are no limitations with respect to the functions that can be used on different units.

High-Speed (HS) parallel usable functionality per unit

	AWG SE X1	AWG SE X2	AWG diff X12	AWG SE X3	AWG SE X4	AWG diff X34	DTZ SE X1	DTZ SE X2	DTZ diff X12	DTZ SE X3	DTZ SE X4	DTZ diff X34	PMU X1	PMU X2	PMU X3	PMU X4
AWG SE X1		n	n	n	n	n	n	y	n	y	y	y	n	n	y	y
AWG SE X2	n		n	n	n	n	y	n	n	y	y	y	n	n	y	y
AWG diff X12	n	n		n	n	n	n	n	n	y	y	y	n	n	y	y
AWG SE X3	n	n	n		n	n	y	y	y	n	y	n	n	n	n	n
AWG SE X4	n	n	n	n		n	y	y	y	y	n	n	n	n	n	n
AWG diff X34	n	n	n	n	n		y	y	y	n	n	n	n	n	n	n

	AWG SE X1	AWG SE X2	AWG diff X12	AWG SE X3	AWG SE X4	AWG diff X34	DTZ SE X1	DTZ SE X2	DTZ diff X12	DTZ SE X3	DTZ SE X4	DTZ diff X34	PMU X1	PMU X2	PMU X3	PMU X4
DTZ SE X1	n	y	n	y	y	y		n	n	n	n	n	n	n	n	n
DTZ SE X2	y	n	n	y	y	y	n		n	n	n	n	n	n	n	n
DTZ diff X12	n	n	n	y	y	y	n	n		n	n	n	n	n	n	n
DTZ SE X3	y	y	y	n	y	n	n	n	n		n	n	y	y	n	n
DTZ SE X4	y	y	y	y	n	n	n	n	n	n		n	y	y	n	n
DTZ diff X34	y	y	y	n	n	n	n	n	n	n	n		y	y	n	n
PMU X1	n	n	n	n	n	n	n	n	n	y	y	y		y	y	y
PMU X2	n	n	n	n	n	n	n	n	n	y	y	y	y		y	y
PMU X3	y	y	y	n	n	n	n	n	n	n	n	n	y	y		y
PMU X4	y	y	y	n	n	n	n	n	n	n	n	n	y	y	y	

High-Resolution (HR) parallel usable functionality per unit

	AWG SE X1	AWG SE X2	AWG diff X12	AWG SE X3	AWG SE X4	AWG diff X34	DTZ SE X1	DTZ SE X2	DTZ diff X12	DTZ SE X3	DTZ SE X4	DTZ diff X34	PMU X1	PMU X2	PMU X3	PMU X4
AWG SE X1		n	n	n	n	n	n	y	n	y	y	y	n	n	y	y
AWG SE X2	n		n	n	n	n	y	n	n	y	y	y	n	n	y	y
AWG diff X12	n	n		n	n	n	n	n	n	y	y	y	n	n	y	y
AWG SE X3	n	n	n		n	n	n	n	n	n	y	n	n	n	n	n

	AWG SE X1	AWG SE X2	AWG diff X12	AWG SE X3	AWG SE X4	AWG diff X34	DTZ SE X1	DTZ SE X2	DTZ diff X12	DTZ SE X3	DTZ SE X4	DTZ diff X34	PMU X1	PMU X2	PMU X3	PMU X4
AWG SE X4	n	n	n	n		n	n	n	n	y	n	n	n	n	n	n
AWG diff X34	n	n	n	n	n		n	n	n	n	n	n	n	n	n	n
DTZ SE X1	n	y	n	n	n	n		n	n	n	n	n	n	n	n	n
DTZ SE X2	y	n	n	n	n	n	n		n	n	n	n	n	n	n	n
DTZ diff X12	n	n	n	n	n	n	n	n		n	n	n	n	n	n	n
DTZ SE X3	y	y	y	n	y	n	n	n	n		n	n	y	y	n	n
DTZ SE X4	y	y	y	y	n	n	n	n	n	n		n	y	y	n	n
DTZ diff X34	y	y	y	n	n	n	n	n	n	n	n		y	y	n	n
PMU X1	n	n	n	n	n	n	n	n	n	y	y	y		y	y	y
PMU X2	n	n	n	n	n	n	n	n	n	y	y	y	y		y	y
PMU X3	y	y	y	n	n	n	n	n	n	n	n	n	y	y		y
PMU X4	y	y	y	n	n	n	n	n	n	n	n	n	y	y	y	

Wave Scale MX PMU specifications

Voltage force

Range	-2.5 V to +6.5 V
Resolution	350 μ V
Accuracy in range -2.5 V to 6.5 V	$\pm(5 \text{ mV} + I \cdot R)$ ^[1]
Accuracy in range -0.5 V to 3.0 V	$\pm(3 \text{ mV} + I \cdot R)$ ^[1]

Voltage measure

Range	-2.5 V to 6.5 V
Resolution	310 μ V
Accuracy in range -2.5 V to 6.5 V	$\pm(5 \text{ mV} + I \cdot R)$ ^[1]
Accuracy in range -0.5 V to 3.0 V	$\pm(3 \text{ mV} + I \cdot R)$ ^[1]

Voltage clamp

Accuracy	$\pm(50 \text{ mV} + I \cdot R)$ ^[1]
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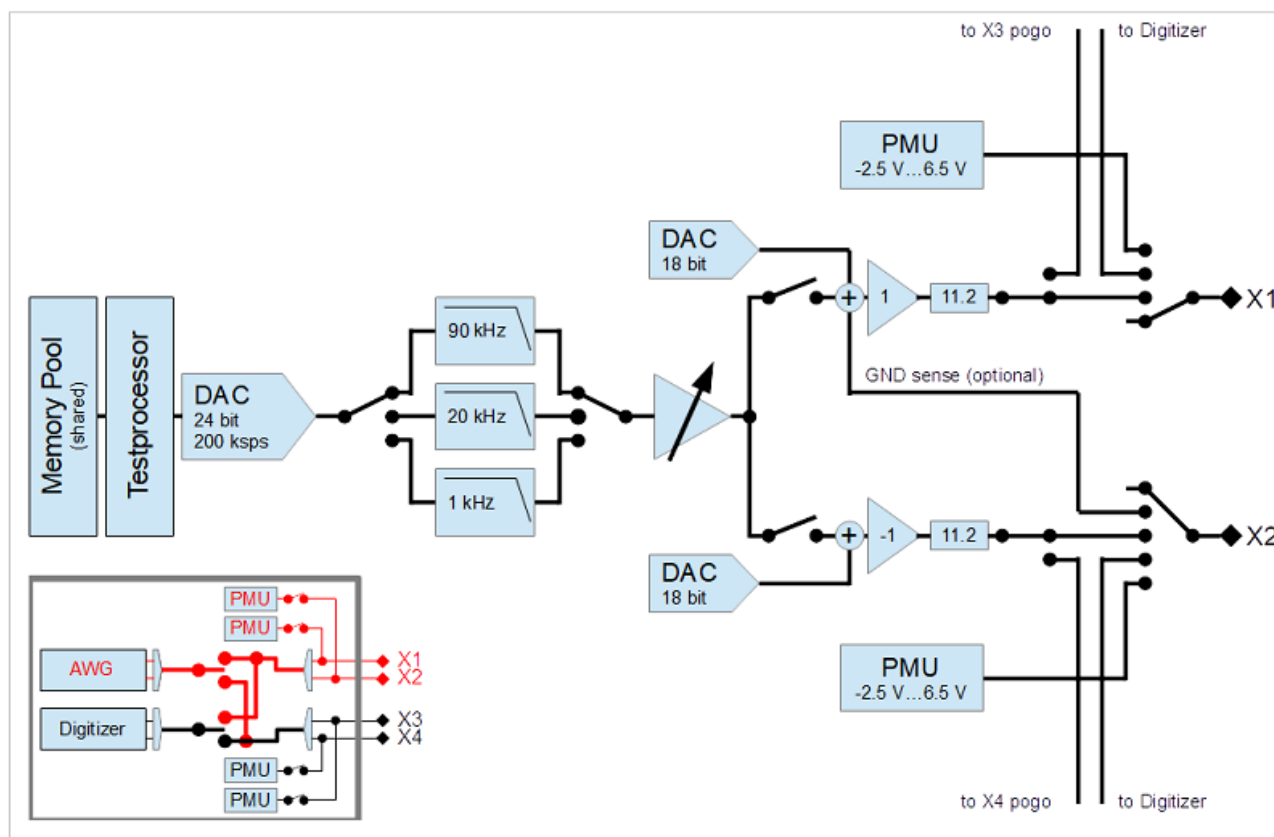
Current force/measure/clamp

Range	Resolution	Measure accuracy	Force accuracy	Minimum clamp distance pos/neg current clamp
$\pm 40 \text{ mA}$ $\pm 20 \text{ mA}$	5 μ A 5 μ A	$\pm 90 \mu\text{A} \pm 0.2\%$ of reading $\pm 90 \mu\text{A} \pm 0.1\%$ of reading	$\pm 90 \mu\text{A} \pm 0.15\%$ of setting $\pm 90 \mu\text{A} \pm 0.1\%$ of setting	$\pm 1.7 \text{ mA}$ $\pm 800 \mu\text{A}$
$\pm 1.9 \text{ mA}$	0.5 μ A	$\pm 2 \mu\text{A} \pm 0.1\%$ of reading	$\pm 2 \mu\text{A} \pm 0.1\%$ of setting	$\pm 50 \mu\text{A}$
$\pm 190 \mu\text{A}$	10 nA	$\pm 280 \text{ nA} \pm 0.1\%$ of reading	$\pm 300 \text{ nA} \pm 0.1\%$ of setting	$\pm 5 \mu\text{A}$
$\pm 19 \mu\text{A}$	1 nA	$\pm 60 \text{ nA} \pm 0.1\%$ of reading	$\pm 60 \text{ nA} \pm 0.15\%$ of setting	$\pm 0.5 \mu\text{A}$
$\pm 4.75 \mu\text{A}$	250 pA	$\pm 30 \text{ nA} \pm 0.2\%$ of reading	$\pm 30 \text{ nA} \pm 0.2\%$ of setting	$\pm 125 \text{ nA}$

^[1] I is the actual current, R is the wiring resistance of $\leq 0.5 \text{ Ohm}$

Wave Scale MX High-Resolution (HR) AWG

Block diagram



WSMX block diagram of the high-resolution AWG

Single-ended signals can be generated with any of the X1, X2, X3 or X4 pogos. On X2 and X4 pogos, signal levels will be inverted. It is not possible to use X1 or X2 simultaneously with X3 or X4. For differential signals, pogo pairs X1 & X2 or X3 & X4 must be used. V_{pos} must be connected to X1/X3, V_{neg} must be connected to X2/X4. The specifications below are only valid if X1/X2 pogos are used. It is not possible to use AWG on X3/X4 and Digitizer on X1/X2 pogos at the same time. Simultaneous usage of AWG and Digitizer is only possible if AWG is connected to X1/X2 and Digitizer to X3/X4 pogos.

Specification	Value
Waveform Memory	144 Msamples shared between 2 consecutive AWG/Digitizer units (unit 1+2, 3+4, ..., 15+16)
Sample Rate	10 ksps to 200 ksps
Resolution	24 bit
Maximum sinewave frequency	90 kHz
Signal Range (single-ended) ^[2]	10.3 Vp @ 10 kΩ to GND

^[2] Voltage level of 10.3 V will be reached for frequencies up to 20 kHz

Specification	Value
DC offset range	-10.3 V to 10.3 V
Output level (AC+DC, single-ended) ^[2]	-10.3 Vp < AC+DC < 10.3 Vp
Output Impedance	X1, X2 pogos: 11.2 Ω $\pm$ 5% X3, X4 pogos: 17 Ω $\pm$ 30%
Selectable hardware filters (0.1dB attenuation frequency, characteristic data)	1 kHz (-3dB @ 1.2 kHz) 20 kHz (-3dB @ 25 kHz) 90 kHz (-3dB @ 106 kHz)
Maximum output current	20 mA
INL (characteristic data)	$\pm$ 3,5 ppm
DNL (characteristic data)	$\pm$ 1,5 ppm
Absolute DC accuracy	$\pm$ (550uV + 0.03% of setting)
Noise Density (typical data)	131 nV/sqrt(Hz) @ 500 Hz, 10 Vp amplitude setting using 1kHz filter 29 nV/sqrt(Hz) @ 20 kHz, 10 Vp amplitude setting using 1kHz filter 18 nV/sqrt(Hz) @ 500 Hz, 0.75 Vp amplitude setting using 1kHz filter 15 nV/sqrt(Hz) @ 20 kHz, 0.75 Vp amplitude setting using 1kHz filter
Short-term drift 0.1 Hz - 10 Hz (peak-to-peak) (characteristic data, differential)	2.2 μ Vpp, 0.75 Vp amplitude setting using 1kHz filter 26 μ Vpp, 10 Vp amplitude setting using 1kHz Filter
RMS noise 22 Hz - 22 kHz (typical data, differential)	2.4 μ Vrms, 0.75 Vp amplitude setting. using 1kHz filter 7.2 μ Vrms, 10 Vp amplitude setting. using 1kHz filter
THD (up to 5th harmonic) 200 k Ω load, differential using THD optimized mode (typical data)	-130 dBc @ 1 kHz, 2Vp -120 dBc @ 1 kHz, 20Vp -117 dBc @ 5 kHz, 2Vp -107 dBc @ 5 kHz, 20Vp -121 dBc @ 20 kHz, 2Vp -104 dBc @ 20 kHz, 20Vp

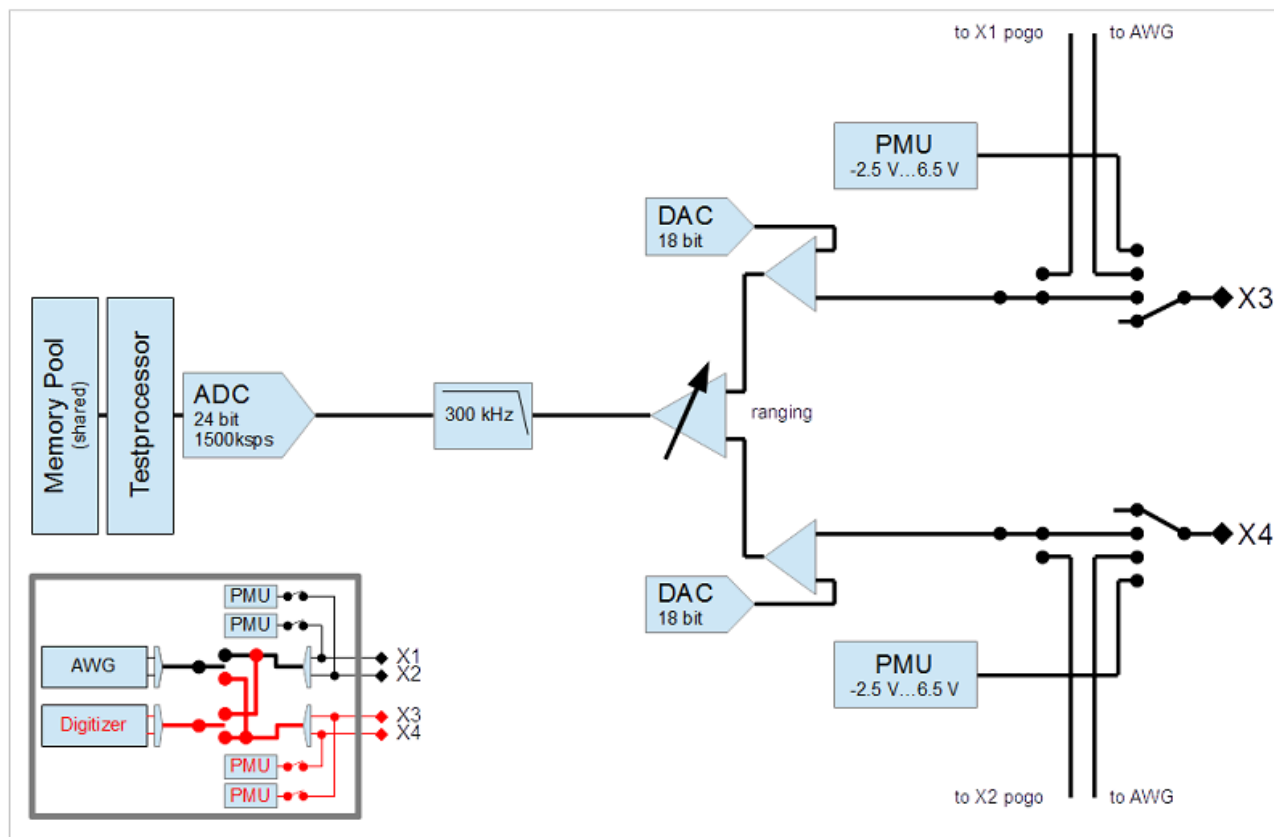
Specification	Value
SFDR ^[3] 200 kΩ load, differential (typical data)	126 dBc @ 1 kHz, 2Vp, 20 kHz bw 121 dBc @ 1 kHz, 20Vp, 20 kHz bw 118 dBc @ 5 kHz, 2Vp, 50 kHz bw 108 dBc @ 5 kHz, 20Vp, 50 kHz bw 123 dBc @ 20 kHz, 2Vp, 130 kHz bw 105 dBc @ 20 kHz, 20Vp, 130 kHz bw
SNR 200 kΩ load, differential (typical data)	114 dBc @ 1 kHz, 2Vp, 20 kHz bw ^[4] 128 dBc @ 1 kHz, 20Vp 20 kHz bw ^[4] 107 dBc @ 5 kHz, 2Vp 50 kHz bw 114 dB @ 5 kHz, 20Vp 50 kHz bw 104 dB @ 20 kHz, 2Vp 130 kHz bw 114 dB @ 20 kHz, 20Vp 130 kHz bw

^[3] SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component including harmonics in the specified frequency range

^[4] EIAJ A-weighted

Wave Scale MX High-Resolution (HR) Digitizer

Block diagram



WSMX block diagram of high-resolution digitizer

Single-ended signals can be connected to any of the X1, X2, X3 or X4 pogos. On X2 and X4 pogos, signal levels will be inverted. For differential signals, pogo pairs X1 & X2 or X3 & X4 must be used. V_{pos} must be connected to X1/X3, V_{neg} must be connected to X2/X4. The specifications below are only valid if X3/X4 pogos are used. It is not possible to use AWG on X3/X4 and Digitizer on X1/X2 pogos at the same time. Simultaneous usage of AWG and Digitizer is only possible if AWG is connected to X1/X2 and Digitizer to X3/X4 pogos.

Specification	Value
Waveform Memory	125 Msamples shared between 2 consecutive AWG/Digitizer units (unit 1+2, 3+4, ..., 15+16)
Sample Rate	1 ksps to 1500 ksps
Resolution	24 bit
Maximum bandwidth	300 kHz ^[5]
Input ranges	10.4 Vp, 5.2 Vp, 2.6 Vp, 1.3 Vp, 520mVp, 260 mVp

^[5] This bandwidth is only usable for signal levels up to -9 dBFS in each range. For full-scale signals, the maximum bandwidth is 100 kHz (max. level at 200 kHz: -6 dBFS)

Specification	Value
Absolute DC accuracy	0.26 Vp range: $\pm (100\mu\text{V} + 0.025\% \text{ of setting})$ 0.52 Vp range: $\pm (100\mu\text{V} + 0.025\% \text{ of setting})$ 1.3 Vp range: $\pm (150\mu\text{V} + 0.025\% \text{ of setting})$ 2.6 Vp range: $\pm (200\mu\text{V} + 0.025\% \text{ of setting})$ 5.2 Vp range: $\pm (300\mu\text{V} + 0.025\% \text{ of setting})$ 10.4 Vp range: $\pm (500\mu\text{V} + 0.025\% \text{ of setting})$
Maximum slew rate of input signal	10.4 Vp range: $6.40 \text{ V}/\mu\text{s}$ 5.2 Vp range: $3.20 \text{ V}/\mu\text{s}$ 2.6 Vp range: $1.60 \text{ V}/\mu\text{s}$ 1.3 Vp range: $0.80 \text{ V}/\mu\text{s}$ 520 mVp range: $320 \text{ mV}/\mu\text{s}$ 260 mVp range: $160 \text{ mV}/\mu\text{s}$
DC offset range	10.4 Vp range: no DC offset 5.2 Vp range: no DC offset 2.6 Vp range: $-5.2\text{V to } +5.2 \text{ V}$ 1.3 Vp range: $-5.2\text{V to } +5.2 \text{ V}$ 520 mVp range: $-5.2\text{V to } +5.2 \text{ V}$ 260 mVp range: $-5.2\text{V to } +5.2 \text{ V}$
Input level (AC+DC)	$-10.1 \text{ V} < \text{AC+DC} < 10.1 \text{ V}$ $-V_{\text{range}} + V_{\text{offset}} \leq V_{\text{diff}} \leq +V_{\text{range}} + V_{\text{offset}}$ $-5.2 \text{ V} + V_{\text{offset}} \leq V_{\text{p}} \leq +V_{\text{offset}} + 5.2 \text{ V}$ $-5.2 \text{ V} - V_{\text{offset}} \leq V_{\text{n}} \leq -V_{\text{offset}} + 5.2 \text{ V}$ $-V_{\text{range}} + V_{\text{offset}} \leq V_{\text{sig}} \leq +V_{\text{range}} + V_{\text{offset}}$
Input bias current (characteristic data)	$\pm 5 \text{ nA}$

Specification	Value
INL (characteristic data)	10.4 Vp range: 2.0 ppm (1.0 ppm with 80 % of range) 5.2 Vp range: 1.6 ppm (0.7 ppm with 80 % of range) 2.6 Vp range: 1.2 ppm (0.7 ppm with 80 % of range) 1.3 Vp range: 1.0 ppm (0.7 ppm with 80 % of range) 520 mVp range: 1.0 ppm (0.8 ppm with 80 % of range) 260 mVp range: 2.5 ppm (1.2 ppm with 80 % of range)
Noise Density w/o DC offset, differential (typical data)	10.4Vp range: 144 nV/sqrt(Hz) @ 500Hz 157 nV/sqrt(Hz) @ 20kHz 5.2Vp range: 81 nV/sqrt(Hz) @ 500Hz 82 nV/sqrt(Hz) @ 20kHz 2.6Vp range: 36 nV/sqrt(Hz) @ 500Hz 39 nV/sqrt(Hz) @ 20kHz 1.3Vp range: 20 nV/sqrt(Hz) @ 500Hz 22 nV/sqrt(Hz) @ 20kHz 520mVp range: 9.8 nV/sqrt(Hz) @ 500Hz 10.5 nV/sqrt(Hz) @ 20kHz 260mVp range: 7.8 nV/sqrt(Hz) @ 500Hz 8.2 nV/sqrt(Hz) @ 20kHz
Short-term drift 0.1 Hz - 10 Hz (peak-to-peak), differential (characteristic data)	10.4 Vp range: 6.6 μ Vpp 5.2 Vp range: 3.9 μ Vpp 2.6 Vp range: 1.7 μ Vpp 1.3 Vp range: 1.1 μ Vpp 520 mVp range: 670 nVpp 260 mVp range: 620 nVpp

Specification	Value
RMS noise 22 Hz - 22 kHz, w/o DC offset, differential (typical data)	10.4 Vp range: 21.3 μ Vrms 5.2 Vp range: 12.6 μ Vrms
THD (up to 5th harmonic) ^[6] w/o DC offset, differential (typical data)	-126 dBc @ 1 kHz, 2 Vp -127 dBc @ 1 kHz, 10 Vp -116 dBc @ 5 kHz, 2 Vp -119 dBc @ 5 kHz, 10 Vp -104 dBc @ 20 kHz, 2 Vp -105 dBc @ 20 kHz, 10 Vp
SFDR ^[7] w/o DC offset, differential (typical data)	116 dBc @ 1 kHz, 2Vp, 20 kHz bw 118 dBc @ 1 kHz, 10 Vp, 20kHz bw 117 dBc @ 5 kHz, 2 Vp, 50kHz bw 118 dBc @ 5 kHz, 10 Vp, 50kHz bw 107 dBc @ 20 kHz, 2 Vp, 300kHz bw 107 dBc @ 20 kHz, 10 Vp, 300kHz bw
SNR w/o DC offset, differential (typical data)	110 dB @ 1 kHz, 2 Vp 20 kHz bw ^[6] ^[8] 119 dB @ 1 kHz, 2 Vp 20 kHz bw ^[6] ^[9] 112 dB @ 1 kHz, 10 Vp 20 kHz bw ^[6] ^[8] 133 dB @ 1 kHz, 10 Vp 20 kHz bw ^[6] ^[9] 105 dB @ 5 kHz, 2 Vp 50 kHz bw ^[8] 116 dB @ 5 kHz, 2 Vp 50 kHz bw ^[9] 107 dB @ 5 kHz, 10 Vp 50 kHz bw ^[8] 130 dB @ 5 kHz, 10 Vp 50 kHz bw ^[9] 98 dB @ 20 kHz, 2 Vp 300 kHz bw ^[8] 109 dB @ 20 kHz, 2 Vp 300 kHz bw ^[9] 100 dB @ 20 kHz, 10 Vp 300 kHz bw ^[8] 123 dB @ 20 kHz, 10 Vp 300 kHz bw ^[9]

^[6] EIAJ A-weighted

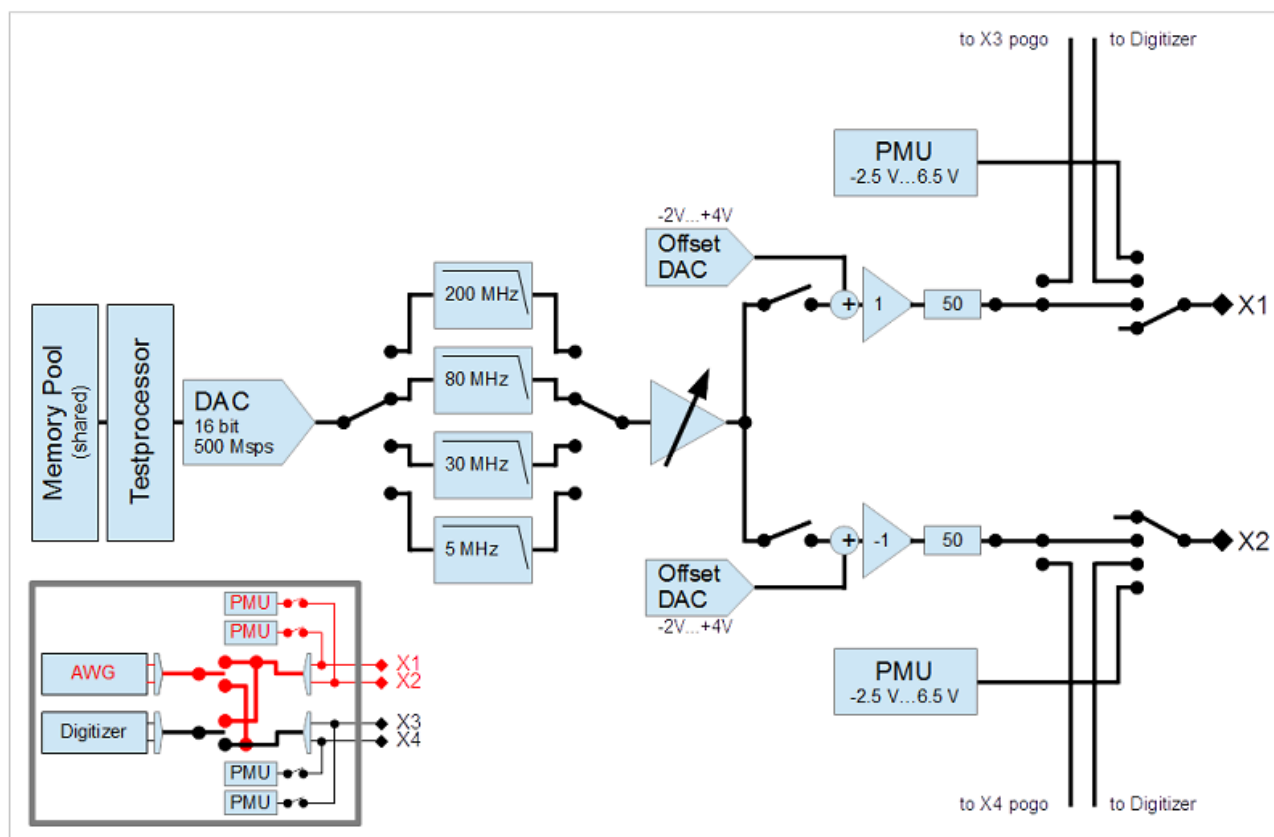
^[7] SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component including harmonics in the specified frequency range

^[8] SNR is measured with a -60dB signal in the Wave Scale MX digitizer range that fits to the full-scale signal and then referred to the full-scale signal. This measurement technique ensures that the distortion components do not affect the noise measurement and is proposed by the Audio Engineering Society, AES17-1998.

^[9] SNR is measured with a -60dB signal in the smallest possible Wave Scale MX digitizer range and then referred to the full-scale signal. This measurement technique ensures that the distortion components do not affect the noise measurement and is proposed by the Audio Engineering Society, AES17-1998.

Wave Scale MX High-Speed (HS) AWG

Block diagram



WSMX block diagram of high-Speed AWG

Specification	Value
Memory	216 Msamples shared between 2 consecutive AWG/Digitizer units (unit 1+2, 3+4, ..., 15+16)
Sample Rate ^[10]	10 Msps to 500 Msps
Resolution	16 bit
Max. sinewave frequency	220 MHz (-3dB attenuation frequency)
Amplitude range	80 mVpp to 5 Vpp single-ended into open 40 mVpp to 2.5 Vpp single-ended into 50 Ω additional attenuation is handled in the system software by downscaling of the waveform
DC offset range	-2 V to 4 V (into open)
Output level (AC+DC)	-2 V < AC+DC < 4 V (into open)

^[10] Lower sample rates are automatically resampled by SmarTest to a range supported by the hardware.

Specification	Value
Absolute DC accuracy	$\pm$ (0.8% of setting +8mV +0.2% of DC offset) (into open)
Maximum output current	40 mA @ 50 Ω (characteristic)
Output DC impedance	50 $\Omega \pm 10\%$
Analog reconstruction filters (0.1dB attenuation frequency)	5 MHz (-3dB @ 5.7MHz) 30 MHz (-3dB @ 37MHz) 80 MHz (-3dB @ 88MHz) 200 MHz (-3dB @ 220MHz)
Noise density @ 10 MHz (typical data)	-141 dBm/Hz for signal amplitudes <1.75Vpp into open
THD (up to 5th harmonic) ^[11] 50 Ω AC termination, 0 dBm Signal 1 V DC offset, single-ended (typical data) 50 Ω AC termination, -10 dBm Signal 1 V DC offset, single-ended (typical data)	-80 dBc @ 5 MHz (5 MHz filter) -78 dBc @ 10 MHz (30 MHz filter) -73 dBc @ 20 MHz (30 MHz filter) -65 dBc @ 50 MHz (80 MHz filter) -52 dBc @ 100 MHz (200 MHz filter) -90 dBc @ 5 MHz (5 MHz filter) -82 dBc @ 10 MHz (30 MHz filter) -84 dBc @ 20 MHz (30 MHz filter) -75 dBc @ 50 MHz (80 MHz filter) -61 dBc @ 100 MHz (200 MHz filter)
SFDR ^{[11][12]} 50 Ω AC termination, 0 dBm Signal 1 V DC offset, single-ended (typical data) 50 Ω AC termination, -10 dBm Signal 1 V DC offset, single-ended (typical data)	81 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 79 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 74 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 66 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 54 dB @ 100 MHz (200 MHz filter, 250 MHz bw) 75 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 81 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 72 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 70 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 60 dB @ 100 MHz (200 MHz filter, 250 MHz bw)

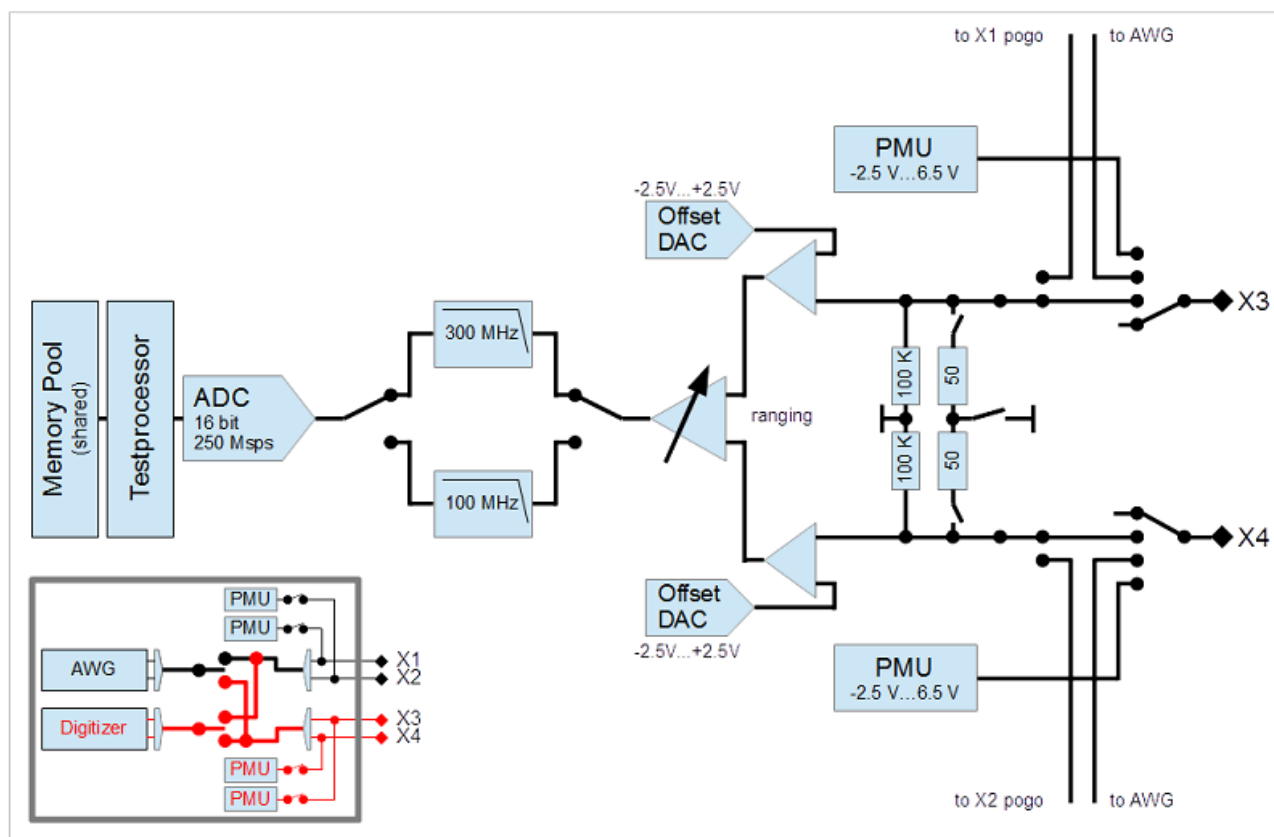
^[11] AC characteristics are measured using signals created by the waveform generation algorithm for sinusoidal signals embedded into SmartTest

^[12] SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component including harmonics in the specified frequency range

Specification	Value
SNR ^[11] 50 Ω AC termination, 0 dBm Signal 1 V DC offset, single-ended (typical data)	67 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 64 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 61 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 57 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 54 dB @ 100 MHz (200 MHz filter, 250 MHz bw)
50 Ω AC termination, -10 dBm Signal 1 V DC offset, single-ended (typical data)	58 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 54 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 51 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 48 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 48 dB @ 100 MHz (200 MHz filter, 250 MHz bw)
SINAD ^[11] 50 Ω AC termination, 0 dBm Signal 1 V DC offset, single-ended (typical data)	67 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 64 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 61 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 55 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 52 dB @ 100 MHz (200 MHz filter, 250 MHz bw)
50 Ω AC termination, -10 dBm Signal, 1 V DC offset, single-ended (typical data)	57 dB @ 5 MHz (5 MHz filter, 25 MHz bw) 54 dB @ 10 MHz (30 MHz filter, 50 MHz bw) 51 dB @ 20 MHz (30 MHz filter, 100 MHz bw) 48 dB @ 50 MHz (80 MHz filter, 250 MHz bw) 48 dB @ 100 MHz (200 MHz filter, 250 MHz bw)
Dual-tone 3rd order intermodulation distortion 50 Ω AC termination, -6 dBm Signal each tone 1 V DC offset, single-ended (typical data)	-79 dBc @ 1 MHz + 1.1 MHz -79 dBc @ 10 MHz + 11 MHz -69 dBc @ 75 MHz + 80 MHz

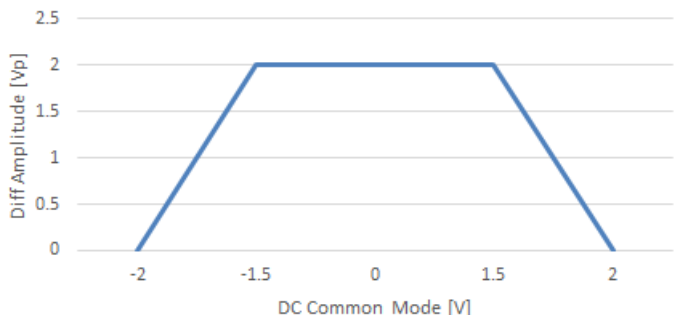
Wave Scale MX High-Speed (HS) Digitizer

Block diagram



WSMX block diagram of the high-speed digitizer

Specification	Value
Memory	216 Msamples shared between 2 consecutive AWG/Digitizer units (unit 1+2, 3+4, ..., 15+16)
Sample Rate	6.25 Msps to 250 Msps
Resolution	16 bit
Maximum input bandwidth	300 MHz
DC offset range (single ended)	-2.5 V ... 2.5 V

Specification	Value
DC common mode (differential) The combination of DC common mode and the amplitude of a differential signal must remain within the area below the curve otherwise clipping will occur.	
Input range (single ended)	62.5mVpp ... 3Vpp
Compliance Range (maximum external voltage, connected state)	-5V to +5V
Absolute DC accuracy	For signal levels (<code>expectedAmplitude</code> property) < 355mVp: $\pm(0.5\% \text{ of reading} + 0.75\% \text{ of offset} + 5\text{mV}) @ 100\text{k}\Omega$ input impedance For signal levels (<code>expectedAmplitude</code> property) > 355mVp: $\pm(1.125\% \text{ of reading} + 0.75\% \text{ of offset} + 7.5\text{mV}) @ 100\text{k}\Omega$ input impedance
Input DC impedance	Single-ended: $50\Omega \pm 10\%$ $100\text{k}\Omega \pm 10\%$ Differential: $100\Omega \pm 5\%$ $100\text{k}\Omega \pm 10\%$ to center ground
Analog Filters	100 MHz 300 MHz
Noise density (typical data)	142 dBm/Hz for expected amplitude setting of 100mVp 141 dBm/Hz for expected amplitude setting of 200mVp 139 dBm/Hz for expected amplitude setting of 300mVp 130 dBm/Hz for expected amplitude setting of 500mVp

Specification	Value
THD (up to 5th harmonic) 50 Ω termination, 250mVp input signal, w/o DC offset, single-ended (typical data)	-79 dBc @ 5 MHz -79 dBc @ 10 MHz -67 dBc @ 50 MHz -59 dBc @ 80 MHz -81 dBc @ 5 MHz -80 dBc @ 10 MHz -75 dBc @ 50 MHz -67 dBc @ 80 MHz
SFDR ^[13] 50 Ω termination, 250mVp input signal w/o DC offset, single-ended (typical data)	80 dBc @ 5 MHz 79 dBc @ 10 MHz 67 dBc @ 50 MHz 60 dBc @ 80 MHz 79 dBc @ 5 MHz 78 dBc @ 10 MHz 72 dBc @ 50 MHz 66 dBc @ 80 MHz
SNR 50 Ω termination, 250mVp input signal, w/o offset, single-ended (typical data)	63 dB @ 5MHz (15MHz bw, 100MHz filter) 58 dB @ 10MHz (50MHz bw, 100MHz filter) 53 dB @ 50MHz (100MHz bw, 100MHz filter) 49 dB @ 80MHz (300MHz bw, 300MHz filter)
50 Ω termination, 75mVp input signal, w/o offset, single-ended (typical data)	55 dB @ 5MHz (15MHz bw, 100MHz filter) 50 dB @ 10MHz (50MHz bw, 100MHz filter) 44 dB @ 50MHz (100MHz bw, 100MHz filter) 41 dB @ 80MHz (300MHz bw, 300MHz filter)

^[13] SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component including harmonics in the specified frequency range

Specification	Value
SINAD 50Ω termination, 250mVp input signal w/o offset, single-ended (typical data)	63 dB @ 5MHz (15MHz bw, 100MHz filter) 58 dB @ 10MHz (50MHz bw, 100MHz filter) 53 dB @ 50MHz (100MHz bw, 100MHz filter) 49 dB @ 80MHz (300MHz bw, 300MHz filter)
50Ω termination, 75mVp input signal w/o offset, single-ended (typical data)	55 dB @ 5MHz (10MHz bw, 100MHz filter) 50 dB @ 10MHz (50MHz bw, 100MHz filter) 44 db @ 50MHz (100MHz bw, 100MHz filter) 41 dB @ 80MHz (300MHz bw, 300MHz filter)
Dual-tone 3rd order intermodulation Distortion 50 Ω termination, 0 dBm Signal, w/o DC offset (characteristic data)	90 dBc @ 1 MHz + 1.1 MHz 82 dBc @ 10 MHz + 11 MHz